

Manuscript

Layer-1

稿件草稿 HTML 版（基于 docs/manuscript_draft.md）

Self-contained academic HTML with embedded SciencePlots figures (base64).

稿件草稿 / Manuscript draft (Layer-1)

EN title: Clinically constrained preoperative planning of indirect mitral annuloplasty: mapping suture dose to AP-diameter reduction, jet location, and LCx safety on a Layer-1 surrogate

中文工作题: 临床约束下的间接二尖瓣成形术术前规划：在一层代理模型上将缝线剂量映射到前后径缩减、反流束位置与回旋支安全

工作短题: Clinical-dose IMA planner (Layer-1)

Nature-skills axes: `paper_type=methods · journal=generic (CMBBE / MedEngPhys / RSOS-class; not flagship Nature) · framework: docs/paper_framework_nature.md`

One-sentence argument: In preoperative IMA planning for FMR, we show that a Layer-1 surrogate can map suture/bridge shortening onto clinically attainable AP reduction, jet location, and LCx safety via a constrained physics-only planner, supported by Galili Level-0 anchors and seed-42 outputs, without claiming new LHHM/Abaqus FSI or first CS-vs-AP comparison.

数据来源: `python run_pipeline.py --seed 42 --paper`（本稿数字均来自该次可复现运行；非新的 Abaqus/LHHM FSI 结果）

验证层级: Level 0（Galili 锚点）+ Level 1（临床剂量映射 + 扫掠/规划器）。Level 2（患者特异 LHHM / 全 FSI）不在本文范围。

Figures: SciencePlots + Times New Roman, $\text{dpi} \geq 300$ (`analysis/plots.py`).

Abstract（草稿）

English abstract

Background: Preoperative planning for indirect mitral annuloplasty (IMA) often treats computational suture/bridge shortening percentages as if they were anteroposterior (AP) diameter reductions. Galili et al. (RSOS 2022) Living Heart Model (LHHM) results show that IMA-AP 50% suture corresponds to **0%** AP reduction, whereas 70% suture collapses AP by **~58%**—a numerical extreme, not a clinical dose.

Methods: On a reproducible Python Layer-1 surrogate (reduced-order FEA + SPH leak index) we implement three contributions: (C1) map suture/bridge shortening % to clinically attainable AP

reduction (MAVERIC ~14–15%, planning ceiling 20%); (C2) continuous design-space sweep plus constrained grid-search preoperative planner; (C3) IMA-CS distal-landing-zone CS–LCx ≥ 8.6 mm and NiTi alternating strain $\leq 0.4\%$. Optional (C4) compares dual vs single suture at matched AP reduction. Main figures use **physics** regurgitation; YAML anchor blending is restricted to Galili validation case IDs.

Results (seed=42): Under clinical mapping the planner evaluates 36 points and retains 30 feasible designs; it recommends **IMA-AP dual suture 60%** ($\eta=0.30 \rightarrow$ AP reduction **18.0%**, physics regurgitation **0.152%**, jet=central). The matched single-suture setting is jet=mixed with a higher commissural fraction. On default anatomy (baseline CS–LCx 11.0 mm), the best feasible IMA-CS is bridge shortening **20%** (CS–LCx exactly **8.6 mm**). Under Galili mapping, IMA-AP 50% remains 0% AP reduction with physics regurgitation $\sim 0.080\%$; 70% reaches $\sim 58\%$ AP reduction with commissural leak.

Conclusions: A Layer-1 surrogate can encode the clinical AP window, jet location, and LCx safety into a deployable preoperative recommendation. It does not replace patient-specific LHHM/FSI and must not treat η as a newly identified FEA parameter.

Keywords: functional mitral regurgitation; indirect mitral annuloplasty; preoperative planning; coronary sinus; LCx; Layer-1 surrogate

中文摘要

背景： 间接二尖瓣成形（IMA）术前规划常把计算文献中的缝线/桥缩短百分比直接当作前后径（AP）缩减百分比；Galili 等 RSOS 2022 的 LHHM 显示 IMA-AP 50% 缝线对应 **0%** AP 缩减，70% 才塌缩至约 **58%** AP 缩减——后者是数值极端而非临床剂量。

方法： 在可复现的 Python 一层代理（降阶 FEA + SPH 泄漏指数）上实现三项贡献：（C1）将缝线/桥缩短 % 映射到临床可达 AP 缩减（MAVERIC ~14–15%，规划上限 20%）；（C2）连续设计空间扫掠 + 约束网格搜索术前规划器；（C3）IMA-CS 远端着陆区 CS–LCx ≥ 8.6 mm 与 NiTi 交变应变 $\leq 0.4\%$ 。可选（C4）双缝线 vs 单缝线在相同 AP 缩减下的交界区泄漏对比。主图均使用 **physics** 反流分数；YAML 锚点混合仅用于 Galili 验证病例。

结果（seed=42）： 临床映射下，规划器在 36 个评估点中选出 30 个可行点；推荐 **IMA-AP 双缝线 60%** ($\eta=0.30 \rightarrow$ AP 缩减 **18.0%**，physics 反流 **0.152%**，jet=central)。同剂量单缝线为 jet=mixed、交界区分数更高。IMA-CS 在默认解剖（基线 CS–LCx 11.0 mm）上，可行最优为桥缩短 **20%**（CS–LCx 恰为 **8.6 mm**）。Galili 映射下 IMA-AP 50% 仍为 0% AP 缩减、physics 反流约 0.08%；70% 为 $\sim 58\%$ AP 缩减并呈 commissural 泄漏。

结论： 一层代理可将临床 AP 窗口、射流位置与 LCx 安全写成可部署的术前建议；它不能替代患者特异 LHHM/FSI，也不能把 η 当作新 FEA 辨识结果。

关键词： functional mitral regurgitation; indirect mitral annuloplasty; preoperative planning; coronary sinus; LCx; Layer-1 surrogate

1. Introduction

English Introduction

Functional mitral regurgitation (FMR) arises from an adverse interaction between ventricular remodeling, mitral annular geometry, leaflet tethering, and impaired leaflet coaptation. Indirect mitral annuloplasty (IMA) approaches seek to modify this geometry without direct leaflet repair, but different device concepts act through different anatomical pathways. Coronary-sinus-based annuloplasty, represented here as IMA-CS and exemplified clinically by the Carillon concept, applies a remodeling effect through the coronary sinus and adjacent mitral annular region. An alternative concept considered in this work, IMA-AP, uses a coronary-sinus-to-interatrial-septum (CS-IAS) suture to produce an anteroposterior (AP) cinching effect. These mechanisms share the objective of improving mitral geometry, but they should not be treated as mechanically or clinically equivalent.

Translation between experimental suture shortening and clinically meaningful AP reduction requires particular caution. The Galili configuration provides useful mechanistic information about how a CS-IAS suture can alter mitral geometry, but its reported suture-shortening percentages are not direct clinical AP doses. Under the convention used in the present scaffold, Galili 50% suture shortening corresponds to **0% AP cinch**; consequently, values such as Galili 70% cannot be interpreted as a 70% clinical reduction. By contrast, the approximately **14–15% AP** change reported in the MAVERIC context provides a clinically grounded order-of-magnitude reference for geometric change, while REDUCE-FMR provides additional directional evidence that coronary-sinus-based remodeling can reduce mitral regurgitation. These studies are used as constraints on plausibility and directionality rather than as direct calibration targets for the surrogate.

The present study evaluates a Layer-1 IMA planning scaffold implemented as a Python surrogate. It is intentionally **not** a production finite-element model and does not reproduce nonlinear tissue mechanics, contact, leaflet stress, device deployment, or patient-specific coronary-sinus deformation. Instead, it combines simplified physics-informed quantities with configurable YAML-based terms to compare candidate IMA-CS and IMA-AP strategies under explicit geometric and safety assumptions. Particular attention is given to the coronary-sinus–left-circumflex-artery relationship: a CS–LCx

separation of at least **8.6 mm** is retained as a literature-informed constraint, while the default scaffold anatomy uses **11 mm**. These quantities function as planning constraints within the surrogate and do not constitute a patient-specific procedural safety assessment.

The contributions are deliberately limited to a transparent pre-FEA planning layer. **C1** maps suture/bridge shortening % onto clinically attainable AP reduction. **C2** provides continuous design-space sweep plus constrained preoperative ranking with inspectable physics vs YAML origins. **C3** encodes LCx clearance and NiTi alternating-strain constraints. Optional **C4** compares dual vs single suture at matched AP reduction. For seed 42, the retained reference planner result is dual IMA-AP **60%**, modeled AP reduction **18%**, physics-channel regurgitation **0.152%**, and jet=central. These are internal surrogate outputs and are not clinical effect sizes.

中文引言要点

- FMR 与 IMA-CS (Carillon 类) / IMA-AP (CS-IAS 缝线) 机制差异, 不可机械等价。
- Galili 计算最优 $\neq$ 临床 AP 剂量; MAVERIC AP 约 14–15%; REDUCE-FMR 仅方向性支持。
- 本文贡献 C1–C3 (及可选 D): 见 docs/paper_plan.md; Level-1 代理, 非生产 FEA。

2. Methods

2.1 English Methods

This study uses a reproducible **Layer-1 Python surrogate** for preoperative IMA design planning. It is **not** a new patient-specific LHHM / Abaqus fluid–structure interaction (FSI) campaign (validation Level 2 remains out of scope; export hooks only).

Pathology and geometry. Posterior papillary pathology is applied as a fractional passive-element state change on a parametric papillary mesh (models/pathology.py). Diastolic baseline annulus circumference (118.5 mm) and AP diameter (34.4 mm) come from the Galili-inspired parametric heart geometry (models/heart_geometry.py).

Device kinematics and clinical dose mapping (C1). Discrete Galili YAML cases and continuous sweep points apply IMA-CS (NiTi bridge shortening) or IMA-AP (CS-IAS suture shortening) through models/devices.py. Two AP mappings are available: (i) **Galili** — reproduces published LHHM geometry (IMA-AP 50% suture $\rightarrow$ 0% AP reduction; 70% $\rightarrow$ ~58% AP / 14.3 mm); (ii) **clinical** — planning transfer

$\backslash\mathrm{AP_reduction\%}=\eta\backslash\times\mathrm{shortening\%},\backslash]$

with IMA-AP $\backslash(\eta=0.30\backslash)$ (50% suture $\rightarrow$ 15% AP) and IMA-CS $\backslash(\eta\approx0.668\backslash)$ (22% bridge $\rightarrow$ $\sim$ 14.7% AP), documented in `configs/design_space.yaml` and `results/clinical_references.yaml`. η is a **planning assumption**, not an imaging-FEA identification. The default planning ceiling is **20%** AP reduction (MAVERIC-attested window $\sim$ 14–15%).

Reduced-order FEA, ROA, and SPH leak index. Peak-systolic loading drives a reduced-order FEA surrogate relating annulus tightening, pathology severity, and coaptation gap to strain/contact (`simulation/run_case.py`, `simulation/loading.py`). Regurgitant orifice area (ROA) is estimated from coaptation gap with optional soft contact-cluster blending for discrete pipeline reporting (`simulation/roa_surrogate.py`). An SPH-inspired leak index ($\sim$ 29k particles honored as a count, not a full Lagrangian SPH solver) converts ROA and gap into **physics** regurgitation (`sph/hemodynamics.py`). At Galili validation case IDs only (pathology, ima_cs_22, ima_ap_50, ...), reported regurgitation may blend physics with YAML anchors (`configs/surrogate_calibration.yaml`); **paper sweep/planner figures use physics only**.

Jet location. Surrogate ROA is split into central vs commissural shares and classified as central / commissural / mixed (`analysis/jet.py`). This is a mechanism label on the surrogate orifice, not echocardiographic PISA/Doppler.

Design sweep and constrained planner (C2–C3). Default grids: IMA-AP 10–70% step 5%; IMA-CS 10–25% step 2%; optional dual-suture IMA-AP on the same AP grid (`analysis/design_sweep.py`). The planner minimizes physics regurgitation subject to AP reduction $\leq 20\%$, NiTi alternating strain $\leq 0.4\%$, and IMA-CS CS–LCx ≥ 8.6 mm (Rottländer 2021), with illustrative baseline CS–LCx = 11.0 mm replaceable by patient CT (`analysis/planner.py`). Reported implants are **grid points**, not interpolated continuous dials.

Optional dual suture (C4). At matched suture % (hence matched clinical AP reduction), dual vs single suture commissural fractions are compared as a mechanism sketch, not a device-clearance study.

η sensitivity. Planner recommendations are re-run at $\eta\pm 20\%$ and exported to `eta_sensitivity.csv` / `.json`. This probes planning-assumption sensitivity, **not** Abaqus/LHHM uncertainty quantification.

Paper artifacts. `python run_pipeline.py --seed 42 --paper` writes five PNGs under `results/output/paper_figures/` and tables under `results/output/paper_tables/` (`analysis/paper_tables.py`, `analysis/plots.py`). Clinical literature alignment tables are

directionality-only (AP↓ / regurg↓); magnitudes are not equated to trial regurgitant-volume percentages.

Honesty. Main manuscript figures report `physics_regurgitation_pct`. This manuscript does **not** invent or report new Abaqus/LHHM cases and must not be cited as production FEA validation.

2.2 模块映射（中文）

步骤	模块
病理 + 参数化瓣环	<code>models/pathology.py</code> , <code>models/heart_geometry.py</code>
装置运动学、Galili vs 临床映射、CS-LCx、双缝线	<code>models/devices.py</code> , <code>configs/design_space.yaml</code>
降阶 FEA（间隙、应变）	<code>simulation/run_case.py</code>
ROA 物理估计	<code>simulation/roa_surrogate.py</code>
SPH 泄漏指数 → physics 反流	<code>sph/hemodynamics.py</code>
射流位置 $\in \{\text{central, commissural, mixed}\}$	<code>analysis/jet.py</code>
连续扫掠	<code>analysis/design_sweep.py</code>
约束规划器（网格搜索）	<code>analysis/planner.py</code>
论文表/图	<code>analysis/paper_tables.py</code> , <code>analysis/plots.py</code>
临床引用与 η 假设	<code>results/clinical_references.yaml</code>
Galili 锚点混合（仅验证病例）	<code>configs/surrogate_calibration.yaml</code> , <code>run_pipeline.py</code>

诚实声明：主文图表使用 `physics_regurgitation_pct`。YAML 混合仅限于病例 ID `pathology / ima_cs_22 / ima_ap_50` 等验证表。本稿不报告、不虚构 Abaqus/LHHM 新算例。

设计空间（默认）：IMA-AP 10–70% step 5%；IMA-CS 10–25% step 2%；双缝线同 AP 网格。约束：AP 缩减 $\leq 20\%$ ；NiTi 交变应变 $\leq 0.4\%$ ；IMA-CS CS-LCx $\geq 8.6\text{ mm}$ （基线解剖默认 11.0 mm，可换患者 CT）。

临床映射假设（Innovation A）：

$$\mathrm{AP_reduction\%} = \eta \times \mathrm{shortening\%}$$
IMA-AP ($\eta=0.30$) (50%

缝线 → 15% AP) ; IMA-CS $\backslash(\eta\approx0.668\backslash)$ (22% 桥 → ~14.7% AP) 。 η 为规划假设, 非新成像-FEA 辨识。

3. Results (叙事; seed=42 实测)

3.1 Galili 锚点复现 (Validation Level 0)

离散 YAML 病例上 (混合报告 vs physics; 见 paper_tables/galili_vs_surrogate.csv) :

Case	Galili 反流 %	Surrogate blended %	Surrogate physics %	Galili AP mm	Galili AP 缩减 %
pathology	5.26	~5.33	~5.34	34.4	0
ima_cs_22	0.29	~0.30	~0.35	34.4	0
ima_ap_50	0.08	~0.09	~0.177	34.4	0

扫掠 (Galili 映射) : IMA-AP 50% → AP 缩减 **0%**, physics 反流 **~0.080%**, jet=central; IMA-AP 70% → AP **~58.4%**, physics 反流 **~11.9%**, jet=commissural (数值极端)。

3.2 临床剂量窗口 vs 数值极端 (C1)

见 clinical_window_vs_numerical_extreme.csv 与 Fig 1-2:

- MAVERIC: 41.4→35.3 mm (**14.7%**)、45.0→38.7 mm (**14%**)。
- 临床映射 IMA-AP 50%: AP **15.0%**, physics 反流 **~0.30%**, jet=central。
- 临床映射 IMA-CS 22%: AP **~14.7%**, physics 反流 **~0.13%**; 在默认 11 mm 基线上 CS-LCx 可能触及 8.6 mm 约束边界 (见规划器)。
- 规划上限: **20%** AP 缩减。

方向性对齐 (非幅度等同) : maveric_reduce_fmr_alignment.csv 将模型在临床窗口内的 AP↓ / 反流↓ 与 MAVERIC (AP↓) 及 REDUCE-FMR (反流↓) 文献方向对照; 不声称百分比幅度可互换。

3.3 约束术前规划器 (C2 + C3)

planner/recommendation.json (clinical 映射) :

- 评估 **36** / 可行 **30**。

- 推荐： IMA-AP 双缝线 **60%**，AP 缩减 **18.0%**（AP 28.21 mm；MAVERIC 标尺 ~33.9 mm），ROA ~9.0 mm²，physics 反流 **0.152%**，jet=central，交界区分数 **0.165**。
- 备选单缝线同 60%： physics **0.160%**，jet=mixed，交界区分数 **0.330**。
- 备选 IMA-CS **20%**： AP 缩减 ~**13.4%**，physics **0.274%**，CS-LCx **8.6 mm**（约束贴边），NiTi 交变应变 **0.34%**（<0.4%）。

$\eta \pm 20\%$ 敏感性见 eta_sensitivity.json/.csv（规划假设扰动，非 FEA 不确定性量化）：

情景	η_{ap}	推荐	AP 缩减 %	physics 反流 %
nominal	0.30	IMA-AP dual 60%	18.0	0.152
-20%	0.24	IMA-AP dual 70%	16.8	0.0895
+20%	0.36	IMA-CS 20%	16.036	0.2293

解读： η 升高使同一缝线 % 更快触顶 20% AP 上限，可行 AP 网格变窄，推荐可跳到可行 IMA-CS； η 降低则允许更高缝线 % 仍落在窗口内。说明规划器对转移效率假设敏感——应在讨论中强调 η 需患者/影像校准，而非当作固定物理常数。

3.4 射流位置与双缝线（C4 / Innovation D）

Fig 3、Fig 5 与 dual_vs_single_matched_ap.csv： 在相同缝线缩短 %（故相同临床 AP 缩减）下比较单/双缝线。

缝线 %	AP 缩减 %	单缝线 jet / 交界分数 / physics %	双缝线 jet / 交界分数 / physics %
50	15.0	central / 0.210 / 0.303	central / 0.105 / 0.298
60	18.0	mixed / 0.330 / 0.160	central / 0.165 / 0.152
70	21.0*	mixed / 0.470 / 0.847	central / 0.235 / 0.347

*70% 临床映射 AP 缩减 21% 超出默认 20% 规划上限，表中作机制对照，不作为推荐植入。

要点： 双缝线在匹配 AP 缩减下降低交界区 ROA 份额，并在 60% 处将 jet 从 mixed 拉回 central；这是机制草图，非器械清关研究。

3.5 LCx / 应变 Pareto（C3）

Fig 4： IMA-CS 临床映射下，反流随桥缩短下降，同时 CS-LCx 逼近 8.6 mm、交变应变逼近 0.4%。默认解剖上可行集以 CS-LCx 约束截断；患者 CT 基线替换后推荐会移动。

4. Figure captions (Fig 1–5)

Fig 1. IMA-AP physics 反流随缝线缩短 % 的变化: Galili AP 映射 (非单调, 70% 恶化) 与临床 $\eta=0.30$ 映射对照; 绿色带为约 14–20% AP 窗口对应的缝线区间 ($\eta=0.30$)。主曲线为 physics, 无 YAML 混合。文件: fig1_ima_ap_nonmonotonic_clinical_window.png。

Fig 2. 装置缩短 % $\rightarrow$ AP 直径缩减 %: Galili LHHM (缝线 % $\neq$ AP %)、临床 IMA-AP ($\eta=0.30$)、临床 IMA-CS ($\eta\approx 0.67$); 水平带为临床 AP 窗口 14–20%; 虚线标 Galili 70% 数值极端 ($\sim 58\%$ AP)。文件: fig2_suture_vs_ap_reduction.png。

Fig 3. 射流位置 / 交界区 ROA 分数随缩短变化 (方=IMA-AP, 圆=IMA-CS); 左右分别为 Galili 与临床映射。水平虚线为分类阈值示意。文件: fig3_jet_location.png。

Fig 4. IMA-CS 临床映射 Pareto 视图: physics 反流 vs CS–LCx (Rottländer 8.6 mm) 及 vs NiTi 交变应变 0.4%。基线 CS–LCx=11.0 mm 为示意解剖。文件: fig4_pareto_lcx_strain.png。

Fig 5. Innovation D: 双缝线 vs 单缝线 IMA-AP 的交界区分数 (及单缝线 physics 反流虚线) 在 Galili / 临床映射下的对照。文件: fig5_dual_vs_single_suture.png。

5. Discussion

English Discussion

The principal interpretive requirement is to keep experimental suture shortening separate from modeled and clinical AP reduction. In the scaffold convention, Galili 50% suture shortening corresponds to **0% AP cinch**. Galili-derived percentages therefore define a mechanical parameterization rather than a clinical dose scale, and a setting such as Galili 70% should not be described as 70% AP reduction or as a 70% therapeutic dose.

The clinical literature instead suggests a substantially narrower geometric context. The approximately **14–15% AP** change associated with MAVERIC provides a useful clinical-scale reference, while REDUCE-FMR supports the direction of coronary-sinus-mediated remodeling. Neither study validates the numerical outputs of the present surrogate. Their appropriate role is to test whether modeled behavior is directionally and geometrically plausible, not to establish quantitative agreement or clinical efficacy.

Coronary anatomy remains an important constraint on IMA-CS strategies. The scaffold retains a CS–LCx separation of at least **8.6 mm** as a literature-informed safety boundary and uses **11 mm** as its default anatomical value. Within that simplified geometry, the CS **20%** candidate lies near the

modeled edge of the admissible region. This finding should be interpreted as a reason for caution within the planner, not as proof that a corresponding intervention is either safe or unsafe in a patient, because the surrogate does not model vessel deformation, device contact, or patient-specific deployment mechanics.

For seed 42, the planner selects the dual-suture AP **60%** candidate, with a modeled AP reduction of **18%**, a physics-channel regurgitation output of **0.152%**, and a central jet classification. The most salient modeled effect of the dual-suture configuration is the change in jet classification; the reduction in the regurgitation surrogate is comparatively mild. The 0.152% value is a model-internal physics output and must **not** be equated with a clinical regurgitant volume, regurgitant fraction, or measured treatment effect.

Interpretation of the selected candidate also requires visibility into the scoring architecture. The planner and validation tables may combine a physics-informed component with YAML-configured blend at Galili case IDs only; paper sweep/planner figures use physics. A plausible final ranking does not by itself demonstrate that the physics component is correct. Reporting physics vs blend separately is important for distinguishing a mechanically motivated result from one driven primarily by priors or heuristic weighting.

Sensitivity to η is a further limitation. A $\pm 20\%$ change in the η assumption can alter the preferred strategy from dual AP 60% to dual AP 70% or CS 20%. The nominal recommendation should therefore be interpreted as conditional on η rather than as a uniquely determined optimum. This instability is scientifically informative: it identifies a parameter whose uncertainty is large enough to change the decision and therefore deserves explicit sensitivity reporting and, where possible, future patient/imaging calibration—not Abaqus/LHHM uncertainty quantification.

Overall, the Layer-1 scaffold is best viewed as a transparent hypothesis-ranking and screening tool preceding higher-fidelity analysis. It can organize competing IMA concepts, enforce simplified anatomical constraints, expose sensitivity, and provide reproducible candidate rankings, but it cannot substitute for patient-specific mechanics, production FEA, benchtop validation, or clinical evidence. The current seed-42 result is consequently a traceable surrogate recommendation rather than a therapeutic prescription.

中文讨论

1. **Galili 50% = 0% AP cinch**: 计算最优不能写成“缩 AP 一半”；临床对话应使用 AP-mm / AP-% (C1) 。
2. **临床 ~15% 窗口**: MAVERIC 对约 14–15%；规划默认上限 20%，避免把 70% Galili 塌缩

当剂量。

- 3. **LCx 8.6 mm:** IMA-CS 必须带入远端着陆区距离；本仓库默认 11→随 cinch 下降，20% 桥处贴边可行。
- 4. **双缝线:** 在匹配 AP 缩减下主要改善交界区泄漏分类，physics 反流降幅相对温和——适合作为机制讨论，而非宣称优越器械。
- 5. 与 **REDUCE-FMR / MAVERIC:** 仅方向对齐（AP↓、反流↓）；幅度不混用。
- 6. **η 敏感性:** $\eta \pm 20\%$ 可改变推荐装置/剂量，说明转移效率需患者/影像校准，而非固定物理常数。
- 7. **定位:** 一层代理是高保真分析前的透明筛选工具，seed-42 是可追溯推荐而非治疗处方。

6. Limitations

- 1. 降阶 FEA + SPH；无在线 Abaqus/LHHM 耦合。
- 2. η 为规划假设，非新成像–FEA 成对辨识。
- 3. 默认 CS–LCx 解剖（11 mm）为示意；阈值 8.6 mm + 患者 CT 才是临床输入。
- 4. 射流分类拆分代理 ROA，非超声 PISA/多普勒。
- 5. 网格搜索报告步进植入设置，不发明连续“旋钮”。
- 6. 双缝线系列为机制草图（Innovation D）。

7. Validation levels

Level	含义	本仓库状态
0	复现 Galili 发表锚点（反流、ROA、几何趋势）	已实现；pytest
1	扫掠 + 临床剂量映射 + 规划约束 + 论文图	已实现；physics 主图
2	患者特异 LHHM / 全 FSI / Abaqus UMAT / 真 29k SPH	超出范围；仅有 fea_export 钩子

勿将 Level-1 曲线表述为 Level-2 FEA。

8. How to regenerate

```
pip install -r requirements.txt # includes SciencePlots
python run_pipeline.py --seed 42 --paper --no-export
$env:PYTEST_DISABLE_PLUGIN_AUTOLOAD=1; python -m pytest tests/ -q
```

图表输出：results/output/paper_figures/（SciencePlots + Times New Roman，dpi $\geq$ 300；该目录在 results/output/ 下默认被 gitignore；需本地/CI 再生）。

写作框架：docs/paper_framework_nature.md（nature-writing methods 轴；勿声称 first IMA-CS vs AP 比较）。

Golden regressions：tests/test_golden_regressions.py（planner seed-42 元组、Galili 50% $\rightarrow$ 0% AP、CS-LCx/ η 守卫）。

Embedded SciencePlots figures

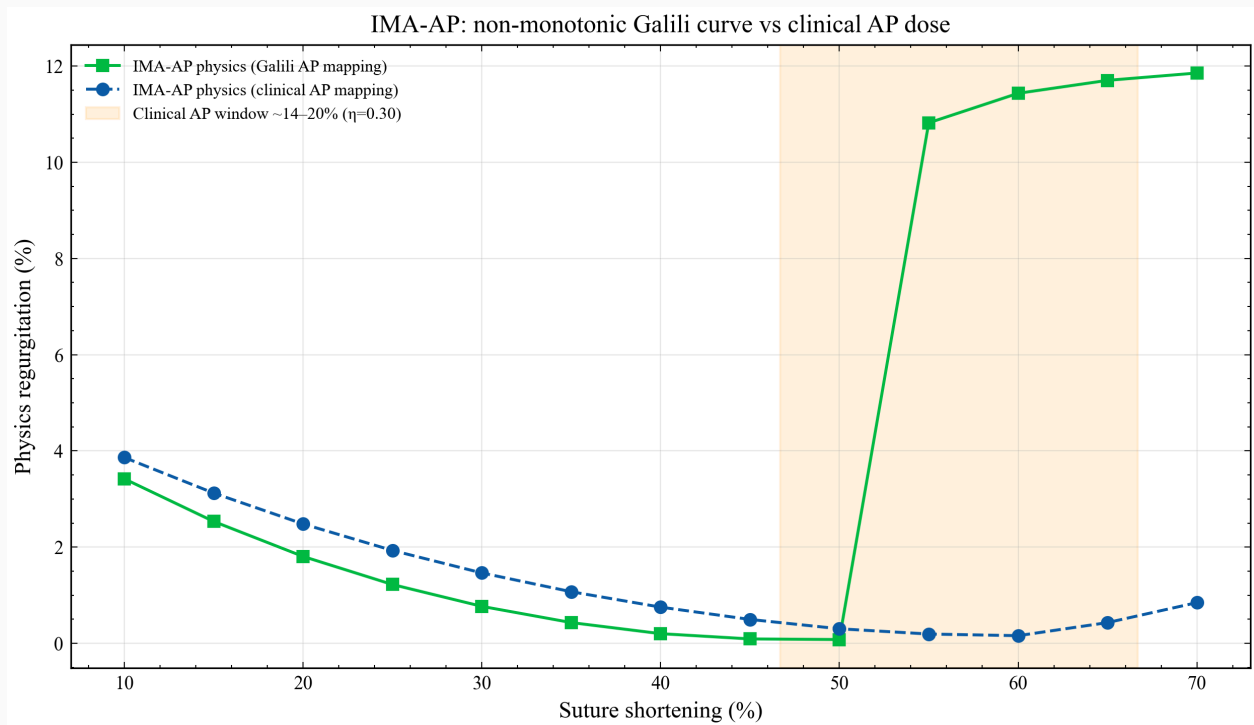


图 1. IMA-AP 物理反流随缝线缩短百分比的变化：Galili 映射 vs 临床映射

文件：fig1_ima_ap_nonmonotonic_clinical_window.png（SciencePlots + Times New Roman, dpi≥300；坐标轴为英文标签，释义见下文中文说明）

来龙去脉与读图说明：本图回答“计算缝线缩短百分比能否直接当作临床前后径（AP，anteroposterior，

指二尖瓣环前后方向直径）剂量？”这一核心问题。横轴为 IMA-AP（indirect mitral annuloplasty–anteroposterior，

间接二尖瓣成形之冠状窦–房间隔缝线路径）的缝线缩短百分比（suture shortening %，装置几何缩短的计算参数）；

纵轴为 Layer-1 代理模型输出的 physics_regurgitation_pct（物理通道反流百分比：由降阶 FEA 间隙与 SPH 泄漏指数导出，不是超声测得的反流容积分数）。

如何读子曲线：图中对照两条映射。（1）Galili 映射复现发表 LHHM（Living Heart Human Model，活体人心计算模型）

离散坐标约定：IMA-AP 50% 缝线在表中仍对应舒张期 AP=34.4 mm，即映射 AP 缩减 0%；70% 缝线才塌缩至约 58% AP，

并出现交界区（commissural）泄漏升高——这是数值极端，不是临床可植入剂量。（2）临床映射采用规划假设

$AP_reduction\% = \eta \times shortening\%$ ，默认 $\eta=0.30$ ，使 50% 缝线对应约 15% AP，落入 MAVERIC（Carillon 临床系列）

约 14–15% 的 AP 窗口。绿色带标出约 14–20% AP 窗口在 $\eta=0.30$ 下对应的缝线区间。

结论（仅限本代理）：临床映射下，反流随缝线缩短在窗口内总体下降且射流偏中央；Galili 映射下 70% 点表现为非单调恶化。主曲线均为 physics，无 YAML 锚点混合。勿将纵轴百分比等同于临床试验反流容积。

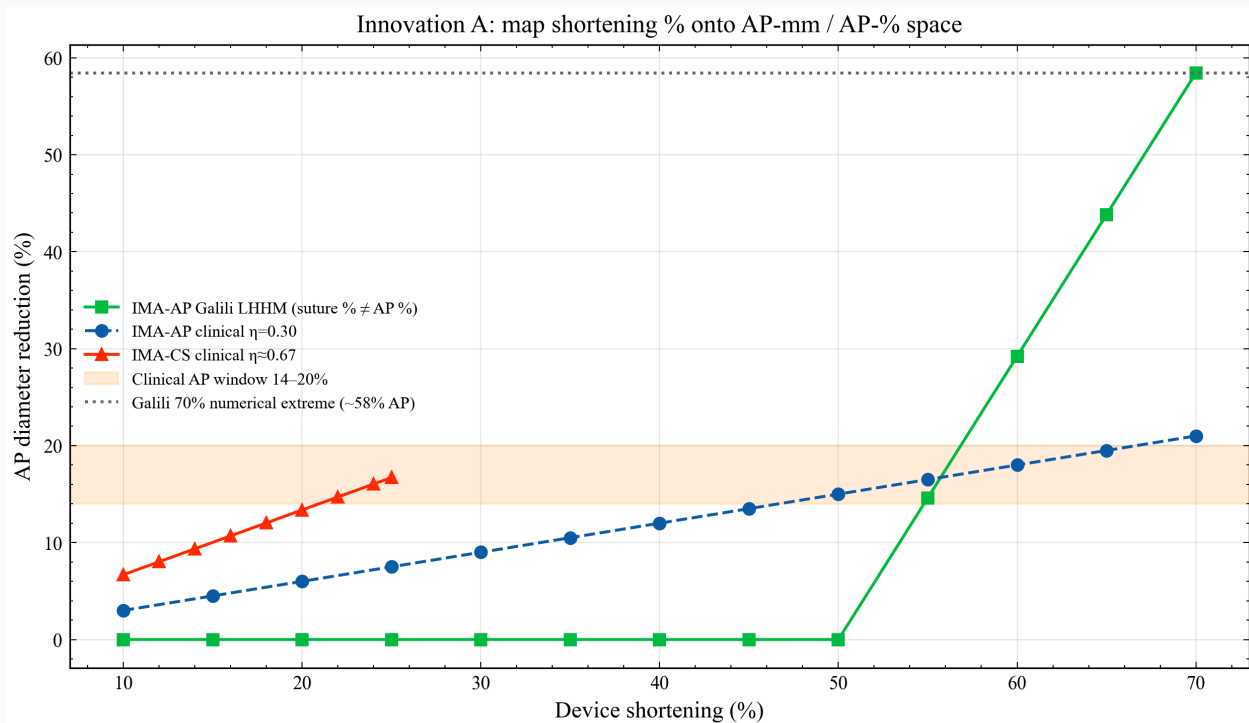


图 2. 装置缩短百分比到 AP 直径缩减百分比的剂量映射

文件: fig2_suture_vs_ap_reduction.png (SciencePlots + Times New Roman, dpi $\geq$ 300; 坐标轴为英文标签, 释义见下文中文说明)

来龙去脉与读图说明: 本图把“计算参数 (缝线/桥缩短 %)”显式翻译成“临床可对话的 AP 直径缩减 %”。

横轴仍为缩短百分比; 纵轴为 AP 缩减百分比。三条 (或多条) 映射轨迹分别对应: Galili LHHM 坐标、

临床 IMA-AP ($\eta=0.30$)、临床 IMA-CS ($\eta\approx0.668$, IMA-CS 为 coronary-sinus-based annuloplasty,

经冠状窦/桥缩短重塑瓣环的路径, 临床代表如 Carillon)。水平带标出临床 AP 窗口 14–20%; 虚线标出

Galili 70% 数值极端 (~58% AP)。

如何读: 若某点落在水平带内, 表示在当前规划假设下几何剂量与 MAVERIC 量级一致; 若落在带外高位,

则属于数值极端, 不应用于术前推荐。本图强调缝线 % $\neq$ AP %: 同一缩短百分比在不同映射下对应截然不同的 AP 缩减。

结论: 规划器默认使用临床映射与 20% AP 上限, 避免把 Galili 70% 塌缩误写成“缩 AP 一半/七成”。

η 是规划假设, 不是新成像-FEA 辨识参数。

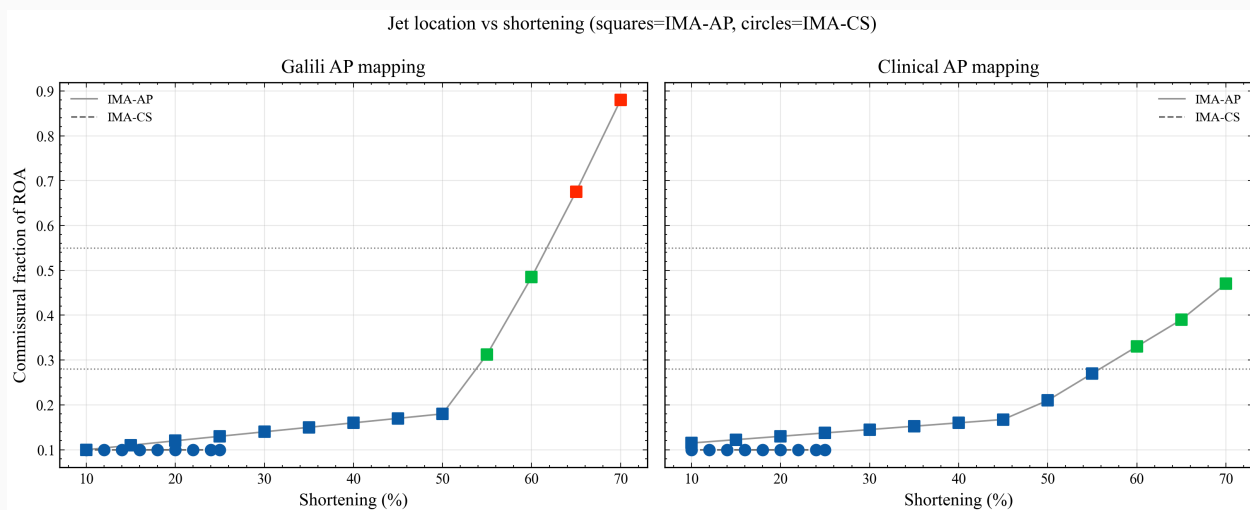


图 3. 射流位置与交界区 ROA 分数随缩短变化（Galili vs 临床）

文件: fig3_jet_location.png（SciencePlots + Times New Roman, dpi≥300; 坐标轴为英文标签, 释义见下文中文说明）

来龙去脉与读图说明：反流“量”之外，规划还关心泄漏机制位置。本代理将 ROA（regurgitant orifice area,

反流口面积，单位 mm^2 ，由对合间隙等代理量估计）拆为中央份额与交界区份额，并分类为 central /

commissural / mixed。横轴为缩短 %；纵轴为交界区 ROA 分数；方块=IMA-AP，圆点=IMA-CS；左右面板对照

Galili 与临床映射。水平虚线为分类阈值示意。

如何读：交界区分数升高意味着泄漏更偏交界区；分类从 central 变为 mixed/commissural 表示机制标签切换。

该标签是代理口上的机制草图，不是超声 PISA（proximal isovelocity surface area）或多普勒诊断。

结论：在临床映射下，IMA-AP 高剂量单缝线更易出现 mixed；IMA-CS 在可行桥缩短范围内多保持 central。

结合图 5，双缝线可在匹配 AP 缩减下压低交界区分数。

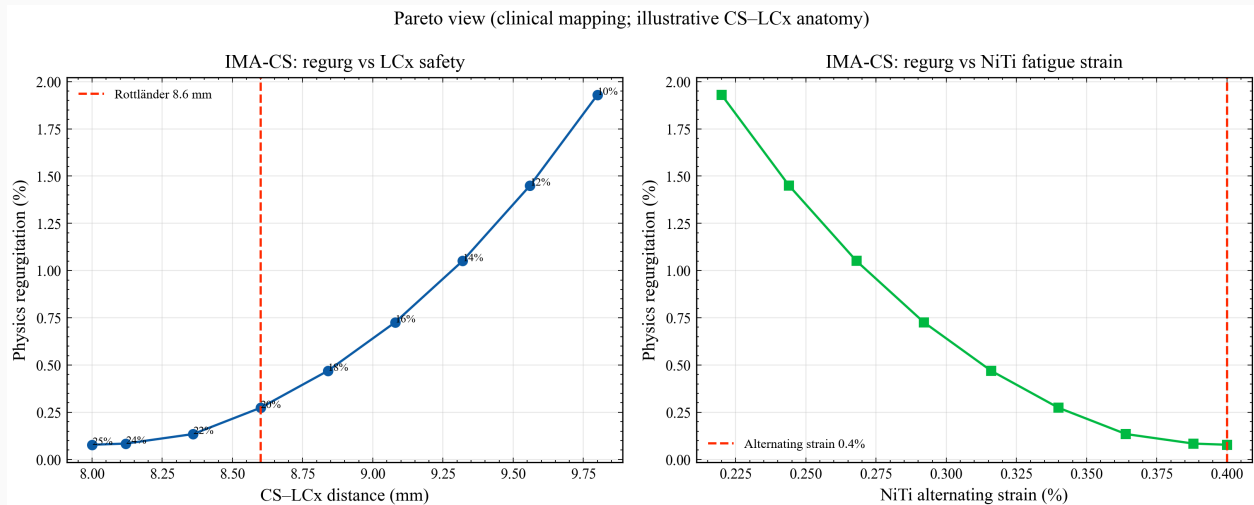


图 4. IMA-CS 临床映射：反流-安全 Pareto（CS-LCx 与 NiTi 交变应变）

文件：fig4_pareto_lcx_strain.png（SciencePlots + Times New Roman，dpi≥300；坐标轴为英文标签，释义见下文中文说明）

来龙去脉与读图说明：IMA-CS 在降低反流的同时可能压缩 CS-LCx 距离（coronary sinus-left circumflex artery，

冠状窦与左回旋支间距）。文献筛查边界取远端着陆区 CS-LCx ≥ 8.6 mm（Rottländer 等；注意：<8.6 mm

预测妥协 ≠ ≥8.6 mm 已证明安全）。同时对 NiTi（镍钛合金桥）交变应变设置 <0.4% 的工程筛查上限。

默认示意解剖基线 CS-LCx=11.0 mm，可替换为患者 CT。

如何读左右面板：横轴多为安全指标（CS-LCx 或交变应变），纵轴为 physics 反流；点随桥缩短移动，

越接近约束边界，可行域越窄。22% 及以上在默认解剖上常因 CS-LCx 违约束而不可行。

结论：默认解剖上可行最优贴近 CS-LCx=8.6 mm 的桥缩短 20%。该结果是规划器筛查信号，不是患者级安全证明。

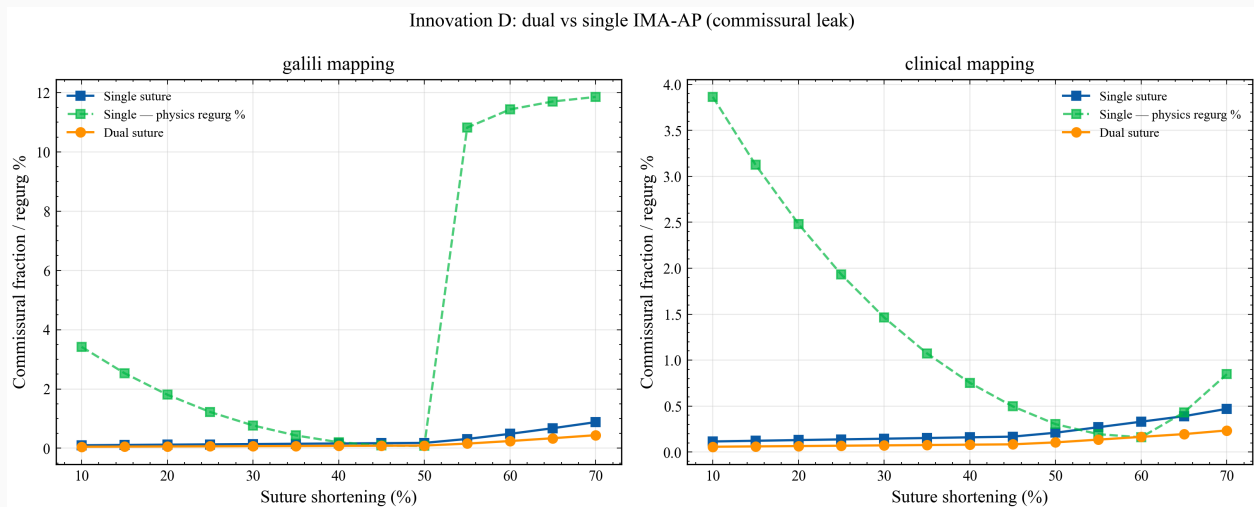


图 5. 双缝线 vs 单缝线 IMA-AP：匹配 AP 缩减下的交界区泄漏对照

文件：fig5_dual_vs_single_suture.png（SciencePlots + Times New Roman，dpi≥300；坐标轴为英文标签，释义见下文中文说明）

来龙去脉与读图说明：可选贡献 C4（Innovation D）在相同缝线缩短 %（故相同临床 AP 缩减）下比较单/双缝线。

纵轴关注交界区 ROA 分数；虚线可叠加单缝线 physics 反流作对照。左右面板对照 Galili / 临床映射。

如何读：在 60% 临床映射点，单缝线 jet=mixed、交界分数 0.330；双缝线 jet=central、交界分数 0.165，

physics 反流由约 0.160% 降至约 0.152%——分类改善更醒目，幅度改善相对温和。70% 点 AP 缩减 21%

超出默认 20% 规划上限，仅作机制对照，不作推荐植入。

结论：双缝线是机制草图，不是器械清关研究；seed-42 规划器推荐双缝线 60% 正是在该对照下选出的网格点。